

Building Graph Representation & Classification

Seagram Building, New York — MaCad Graph Machine Learning Seminar, Assignment 03

Building Overview

The Seagram Building (1958, Mies van der Rohe) was selected as the subject of this assignment. It is a canonical high-rise office tower in New York City, characterised by a clear structural logic: a rectangular tower rising from a podium/plinth level, lifted on columns, with a central service core. The building was modelled in Rhino and exported as five separate OBJ layers: ground, columns, plinth, offices, and core.

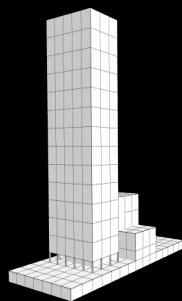


Fig. 1 — CellComplex geometry

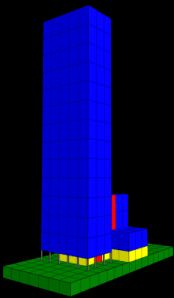


Fig. 2 — CellComplex colored by spatial type

Graph Construction

The CellComplex was converted into a Building Graph Representation (BGR) using TopologicPy. Each spatial cell becomes a graph node; shared faces between cells become edges encoding adjacency. Nodes carry one-hot encoded features (5 categories: ground=0, column=1, plinth=2, office=3, core=4) and are colored by cell type for visualisation.

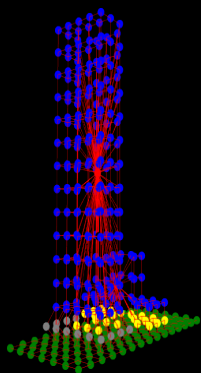


Fig. 3 — BGR adjacency graph (green=ground, gray=columns, yellow=plinth, blue=offices, red=core)

Classification Results

The exported graph CSV was passed to the pre-trained GraphSAGE model (bgr_model.pt). My assessment was **Adherence with Plinth (3)**. The model predicted **Separation with Plinth (1)** with 100% confidence, indicating it interpreted the core as spatially separated from the office cells rather than adhering to them.

Actual Value	Predicted Value	Actual Label	Predicted Label	Confidence
3	1	Adherence with Plinth	Separation with Plinth	1.0

The discrepancy between my label (Adherence) and the model's prediction (Separation) highlights how graph topology encodes spatial relationships differently from human visual assessment. The model's interpretation — that the core is separated from the office mass — is geometrically defensible given the discrete cell boundaries in the simplified massing model.